# Elements of AIML
# Unit 04 – Lecture 01 Notes
# Supervised Learning (Regression)

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# 1 Lecture Overview

This lecture introduces **regression**, a supervised learning task where the target is a continuous value. We discuss linear regression, loss functions, evaluation metrics, and overfitting/regularization.

*Sources:* Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.

# 2 How to use these notes (self-study)

For regression, always separate three things: (1) the model form ($\hat{y} = f(x)$), (2) the loss used for training (e.g., MSE), and (3) the metric used for evaluation (MAE/RMSE/$R^2$). Work through the mini metric computation example and then attempt the practice questions.

# 3 Learning Outcomes

- Define regression and give real-world examples.
- Explain linear regression and the role of a loss function.
- Compute and interpret MAE, RMSE, and $R^2$.
- Explain overfitting and the purpose of regularization.

# 4 Keywords

- **Regression:** predicting a continuous target.
- **Loss function:** objective minimized during training (e.g., MSE).
- **MAE/RMSE:** error metrics in target units.
- **$R^2$:** variance-explained metric (relative measure).
- **Overfitting:** fits training noise; poor generalization.
- **Regularization:** penalty term to control complexity.

# 5 Abbreviations and symbols

- $X$: feature matrix, $y$: true targets, $\hat{y}$: predictions.
- $n$: number of training examples, $d$: number of features.
- $w$: weight vector (parameters), $w_0$: intercept (bias term).
- $e_i = y_i - \hat{y}_i$: residual/error for example $i$.
- MAE/MSE/RMSE: regression metrics; $R^2$: coefficient of determination.
- $\lambda$: regularization strength (Ridge/Lasso).

# 6 Regression basics

In regression, we learn a function $f$ so that:

$$\hat{y} = f(x)$$

where $y$ is continuous (e.g., price). Regression is used in forecasting, estimation, and prediction problems.

## 6.1 Regression vs classification (quick reminder)

- **Regression**: output is a real number (temperature, price).
- **Classification**: output is a class label (spam/ham).

# 7 Linear regression

## 7.1 Model

For $d$ features:

$$\hat{y} = w_0 + \sum_{j=1}^{d} w_j x_j$$

The goal is to find weights $w$ that minimize prediction error on training data.

## 7.2 Vector/matrix form (compact notation)

If we add a constant "1" feature for the intercept, we can write:

$$\hat{y} = Xw$$

where $X \in \mathbb{R}^{n\times(d+1)}$ and $w \in \mathbb{R}^{(d+1)}$. This form is useful for reasoning about optimization and implementation.

## 7.3 Loss functions

A common loss is mean squared error (MSE):

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

RMSE is $\sqrt{\text{MSE}}$ and is easier to interpret because it is in the same units as $y$. MAE uses absolute values:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

## 7.4 MAE vs MSE (why do we have both?)

- MSE/RMSE penalizes large errors strongly (squares the error), so it is sensitive to outliers.
- MAE penalizes errors linearly, often more robust to outliers.
- In practice, choose based on the cost of large mistakes (domain-dependent).

# 8 Evaluation metrics

## 8.1 MAE vs RMSE

- MAE treats all errors linearly.
- RMSE penalizes large errors more strongly.

Choice depends on whether large errors are especially costly.

## 8.2 $R^2$ (coefficient of determination)

One common definition:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$R^2$ is relative to a baseline that predicts the mean $\bar{y}$. It can be misleading when:

- comparing across datasets with different variance,
- evaluating non-linear relationships with a poor model,
- using it without checking residual patterns.

## 8.3 Mini worked example (compute MAE/RMSE)

Suppose:

$$y = [50, 60, 80], \quad \hat{y} = [55, 58, 70]$$

Residuals: $e = y - \hat{y} = [-5, 2, 10]$.

- MAE $= \frac{|-5|+|2|+|10|}{3} = \frac{17}{3} \approx 5.67$
- MSE $= \frac{25+4+100}{3} = 43$, so RMSE $= \sqrt{43} \approx 6.56$

# 9 Underfitting, overfitting, and regularization

## 9.1 Underfitting vs overfitting

- Underfitting: too simple; high error on train and test.
- Overfitting: too complex; low train error, high test error.

Always evaluate using proper splits/cross-validation (Unit III).

## 9.2 Residual analysis (basic sanity check)

Even if metrics look good, we should inspect residuals:

- If residuals show patterns (e.g., curve shape), a linear model may be misspecified.
- If residual spread increases with $x$, variance is not constant (heteroscedasticity).
- Large residuals may indicate outliers or missing features.

### 9.3 Regularization idea

Regularization adds a penalty to control complexity. Example (Ridge):

$$\text{Loss} = \text{MSE} + \lambda \|w\|^2$$

Lasso uses $\|w\|_1$ and can push weights to zero.

### 9.4 Ridge vs Lasso (high-level)

- **Ridge (L2)**: shrinks weights smoothly; helps with multicollinearity.
- **Lasso (L1)**: can set some weights to exactly zero; acts like feature selection.

Regularization reduces variance (overfitting) at the cost of some bias.

## 10 Minimal Python example (illustrative)

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state
    =42)
model = LinearRegression().fit(X_train, y_train)
pred = model.predict(X_test)

mae = mean_absolute_error(y_test, pred)
rmse = np.sqrt(mean_squared_error(y_test, pred))
print(mae, rmse)
```

## 11 Common pitfalls (and how to avoid them)

- **Evaluating on training data:** this hides overfitting; always use proper splits/CV.
- **Ignoring units:** RMSE is in the same units as $y$; MSE is in squared units.
- **Over-trusting $R^2$:** check residuals and absolute error; $R^2$ can be misleading across datasets.
- **Forgetting preprocessing:** outliers, scaling, and feature encoding can dominate performance.

## 12 Practice (with answers)

### P1. When would you prefer MAE over RMSE?

**Answer:** when you want robustness to outliers or when large errors should not dominate the metric too strongly.

### P2. What does $\lambda$ do in Ridge regression?

**Answer:** it controls the penalty strength; larger $\lambda$ shrinks weights more (simpler model), often reducing overfitting.

### P3. Give one reason why $R^2$ can be misleading.

**Answer:** it depends on variance of $y$; a model can have a high $R^2$ but still have large errors in absolute terms, or a low $R^2$ even when RMSE is acceptable.

## 13 Exit questions (with answers)

### Q1. Regression vs classification?

**Answer:** regression predicts a continuous value; classification predicts a discrete class label.

### Q2. Why is RMSE easier to interpret than MSE?

**Answer:** RMSE has the same units as the target variable (MSE is squared units).

### Q3. What does $R^2$ measure and when is it misleading?

**Answer:** $R^2$ measures variance explained relative to predicting the mean; it can be misleading across datasets, with non-linear patterns, or without residual analysis.

### Q4. Overfitting mitigation strategies?

**Answer:** regularization, more data, simpler model, better features, cross-validation, early stopping (for iterative models).

## 14 Summary

Regression is a core supervised learning task. Correct evaluation and controlling overfitting are essential for reliable predictions.

*Sources:* Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.